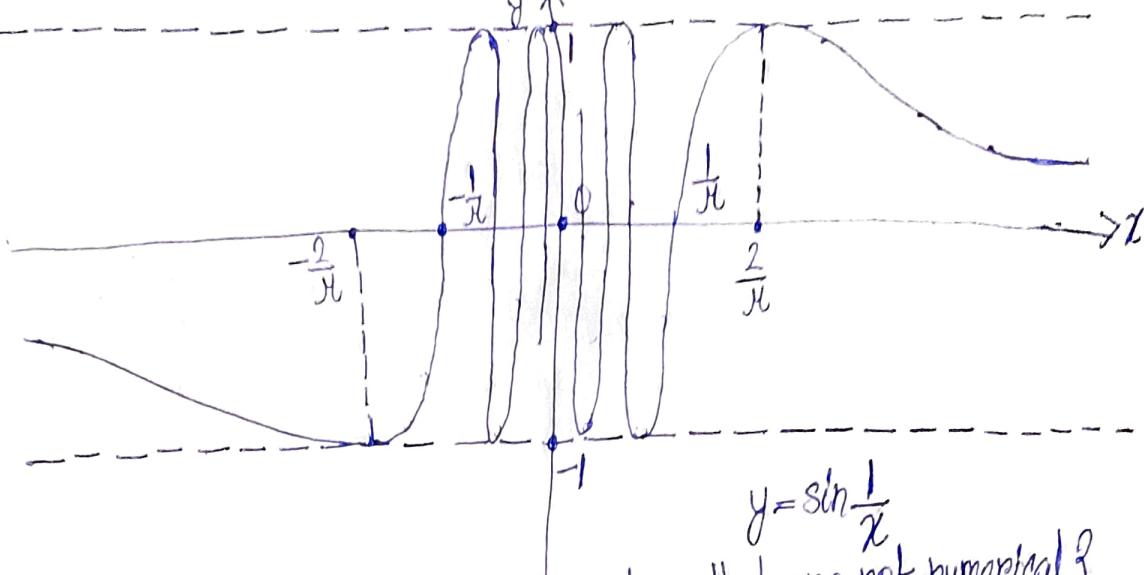




Reader: Does this statement hold for functions that are not numerical?

Author: Yes, it does. In the most general case we can give the following

Definition: The graph of a function f defined on a set D with a range on a set E is a set of all pairs (x, y) such that the first element of the pair x belongs to D , while the second element of the pair y belongs to E , y being a function of x ($y = f(x)$).

Reader: So it turns out that the graph of a function such as the area of a circle is actually a set of pairs each consisting of a circle (an element x) and a positive number (an element y) representing the area of a given circle.

Author: Precisely so. Similarly, the graph of a function representing a schedule of students on duty in a classroom is a set of pairs each containing a date (an element x) and the name of a student (an element y) who is on duty on this date. Note also that in practice this function indeed takes a graphic form.

If in a particular case both elements of the pair (both x and y) are numbers, we arrive at the graph of the function represented by a set of points on the coordinate

Plane. This is the familiar graph of a numerical function.

Prob 8: Find the function $f(x)$ if $f(x-1) - f\left(\frac{1-x}{x}\right) = 2x$. ①

Let, $\frac{1-x}{x} = y$ $\frac{1}{x} = y+1$ $x = \frac{1}{y+1}$ $x-1 = \frac{1}{y+1} - 1 = \frac{1-y}{y+1}$

$f\left(\frac{1-y}{y+1}\right) - f(y) = \frac{2}{y+1}$ (Wrong)